Middle-Rung Law: Predictions of a Three-Rung Frequency Ladder and Formula Compendium

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Normalization relation $4\pi^2r^3c^2/(T^2Ghv)=1$ written by Zhang Xiangqian (2026-03-03). All numerical values use CODATA 2018 constants.

Part I — Preregistration and Falsifiable Predictions

1. Hypothesis

The Planck frequency is the geometric mean of the cosmic expansion rate H and an ether frequency ν_e: at every epoch t, $\nu_e(t) \cdot H(t) = \nu_P^2$, equivalently $\hbar G = c^5/(4\pi^2\,\nu_e\,H)$. ν_e equals the de Broglie angular frequency of the Hubble sphere treated as one object (Kepler's third law with $r = c/H$, $T = 2\pi/H$, combined with $Mc^2 = h\nu$; normalization relation written by Zhang Xiangqian, 2026). Prior to cosmic age 3.912 Gyr (3.641–3.950 Gyr across H_0 = 61.8–63.5 km s^{-1} Mpc^{-1}), the early universe undergoes hot radiation/matter evolution and intense hierarchical merging of smaller structural units; at **9.900 Gyr ago** (range 9.893–9.933 Gyr ago, redshift $z_s = 1.686$, range 1.669–1.817; acoustic-angle benchmark $z_s = 1.721$ gives age 3.837 Gyr and scale ratio $1+z_s = 2.721 \approx e$), large-scale galaxy structures stabilize and cosmic expansion locks into uniform coasting $H(z) = H_0(1+z)$, after which $\nu_e(t)$ rises linearly with cosmic age while H(t) declines inversely with fixed product $\nu_P^2 = 8.715\times10^{84}$ Hz².

2. Predictions

ID	Quantity	Predicted value	Deciding data	Falsified if
P1	Hubble constant H_0	61.8–63.5 km s^{-1} Mpc^{-1}	Gravitational-wave standard sirens (free of astrophysical calibration)	Measured value outside range at 3σ (e.g. 73)
P2	Age of universe	13.5–13.9 Gyr (central 13.82 Gyr)	Oldest stellar ages + independent cosmological determinations	Reliably < 13.4 Gyr or > 14.0 Gyr
P3	Expansion history & coasting transition	Transition to uniform coasting $H(z) = H_0(1+z)$ at 9.900 Gyr ago (9.893–9.933 Gyr ago, age 3.912 Gyr, $z_s \approx 1.69$); at z = 2.33, H is ~3% below ΛCDM	DESI final, Euclid, and Roman BAO measurements	z = 1.5–2.5 consistent with ΛCDM at < 1% precision with no transition
P4	Dark matter	Early compact extreme objects, 10^{14}–10^{20} kg	Microlensing, femtolensing, sub-solar gravitational-wave searches	Mass window completely excluded
P5	MOND-like acceleration scale	$a_0 = (0.96–0.98)\times10^{-10}$ m s^{-2} today, growing with H(z): ×2 at z = 1, ×3 at z = 2	High-redshift disk galaxy rotation curves	No redshift evolution
P6	The Planck gate (non-classical motional limit)	No mechanical degree of freedom with effective mass $m_{eff} \geq m_P$ = 21.8 μg (total frequency ≥ $\nu_P = 2.952\times10^{42}$ Hz) can exist as a wave at the human rung or be prepared in a non-classical motional state	High-overtone bulk acoustic resonators + superconducting qubits, levitated particles, massive optomechanics	A peer-reviewed experiment prepares ≥3σ Wigner negativity or a cat-state superposition (separation > 2× zero-point amplitude) for $m_{eff} \geq$ 21.8 μg
P6.1	The Planck blind band & observational collective effect	Within 21.8 ± 0.3 μg (21.5–22.1 μg baseline, wider under strong collective coupling), field-pressure readout between the target and the observational collective decouples: both classical and non-classical observations vanish (<1% of out-of-band baseline, resuming outside), with window centre and width shifting reproducibly across laboratories, nearby heavy equipment, and observer body-mass frequencies	Tunable bulk-acoustic / optomechanical resonators spanning 16–25 μg, microgram levitated-particle scattering scans, and differential runs across laboratories, heavy equipment, and manned vs. unmanned operation	A peer-reviewed scan across 21.0–22.6 μg observes smooth, continuous classical optical/mechanical scattering and readout throughout the entire interval (<10% attenuation)

Derived quantities: $N = \nu_P/H = (1.44–1.47)\times10^{60}$; $\nu_e = (4.2–4.4)\times10^{102}$ Hz; mid-rung frequency $\sqrt{(H\,\nu_P)}$ = 2.43–2.46 THz. Ether-flow (gravitational potential) variations add in quadrature; under current axioms, a dual-optical-clock comparison next to any device without mass change should read zero.

2.1 Operational Criteria and the Observational Collective Mechanism for P6 and P6.1

Physical mechanism (the Observational Collective): The Planck mass $m_P = \sqrt{(\hbar c/G)} = 21.8$ μg has total frequency $m_P c^2/h$ equal to the human-boundary frequency $\nu_P = 2.952\times10^{42}$ Hz. By Axioms 1, 3, and 4, an observation is never an isolated object acting alone: the target matter (near 21.8 μg), the probing light, the vacuum, the measuring apparatus, the observer's eye and body (whose body mass M_{obs} carries its own ether frequency $F = M_{obs}c^2/h$), and the surrounding laboratory building and nearby heavy equipment are all distinct ether structures with their own frequencies and overlapping ether spaces. Human observation is the field-pressure coupling ($v^2 = 2\Phi$) across this entire **observational collective**. When the effective mass of a target mode enters the neighbourhood of ν_P (21.8 μg), the collective frequency of the coupled system gates human-layer readout at ν_P: field-pressure transduction decouples, causing **both classical and non-classical observations to vanish simultaneously** within the Planck blind band (P6.1); above the upper edge of the blind band (> 22.1 μg), classical field-pressure readout resumes, but because the total frequency exceeds ν_P, the structure can no longer exist as a wave at the human rung and no non-classical motional state can be formed (P6).

Criterion	Fixed operational definition
Effective mass m_{eff}	The effective (motional) mass of the mechanical/vibrational mode as reported by the experiment; for centre-of-mass motion or levitated particles, the total mass of the target.
P6 non-classical state criteria (either suffices)	(i) Wigner-function negativity of the motional state at ≥ 3σ significance; (ii) a superposition of two coherent states (cat state) separated in phase space by more than twice the zero-point amplitude, evidenced by interference fringes or Wigner negativity.
P6.1 dual-channel disappearance criteria (both required)	(i) Inside the target's collective blind window (baseline 21.8 ± 0.3 μg), the effective scattering cross-section or readout response across all photon-, phonon-, or field-pressure-mediated classical (optical scattering, thermal emission, centre-of-mass readout) and non-classical channels drops to **< 1% of its out-of-band baseline (or below instrument noise floor)**; (ii) tuning m_{eff} or measuring control samples outside the window (< 21.5 μg or > 22.1 μg) immediately restores normal classical scattering and mechanical readout (ruling out sample loss or damage).
Why it is an interval & collective environmental shift	Because the decoupling threshold is governed by the superposed collective frequency field of the target, probing light, vacuum, detector, observer's eye and body ($F_{body} = M_{body}c^2/h$), laboratory building, and surrounding heavy apparatus, the blind region is a finite band rather than a mathematical point at 21.764 μg. Repeating the measurement on the same target across **different laboratories, toggling or relocating nearby large-scale equipment (high-field magnets, heavy machinery, high-power RF sources), or varying the presence and body masses (frequencies) of near-field human observers** produces a **reproducible shift in both the centre mass and width of the blind window**.
Non-qualifying cases	For P6: near-ground-state cooling, squeezing, or Gaussian entanglement with everywhere-positive Wigner functions (e.g. near-ground-state cooling of the 10 kg LIGO mirrors is not a counterexample). For P6.1: signal loss from sample evaporation, mechanical detachment, optical misalignment, cavity detuning, or ordinary detector malfunction.
Expected trend (not a falsification condition)	Approaching the collective blind band from either side produces a steeper edge drop in both classical signal-to-noise ratio and non-classical cat-state coherence time than environmental decoherence models predict; the dip window widens when heavy surrounding equipment is active or multiple observers are in close proximity compared to unmanned remote telemetry.

3. Disclosure of Data Already Examined

Experimental and observational datasets examined prior to or upon fixing these predictions:

- **P5:** Ciocan et al. 2026 (*Astronomy & Astrophysics* 709, L16) report a_0 rising with redshift, reaching 2.38×10^{-10} m s^{-2} at $z \approx 1$. The direction agrees with P5; the magnitude is ~19% above our prediction. Earlier analyses (Milgrom 2017; Limbach et al. 2008) disfavour strong evolution.
- **P3:** DESI DR2 BAO data (13 points). Our model gives best $\chi^2 = 27.7$ (fixed acoustic-angle calibration; 21.64 vs. 10.54 for ΛCDM when r_d is free under identical degrees of freedom), which is unfavourable to our model, with the largest residual at $z = 0.934$.
- **P1:** Current gravitational-wave-only standard-siren bounds (±15–20 km s^{-1} Mpc^{-1}) do not yet discriminate.
- **P6 & P6.1:** The heaviest mechanical mode prepared in a non-classical state to date is Bild et al. 2023 (*Science* 380, 274), demonstrating cat states at $m_{eff} = 16.2$ μg (74% of the 21.8 μg Planck gate); macroscopic particle scattering experiments have not yet performed high-resolution continuous mass-stepped coupling scans across the 21.5–22.1 μg window. Standard physics predicts neither a non-classical limit nor a dual-channel blind band at 21.8 μg, making both P6 and P6.1 directly falsifiable within months to years.

Every future comparison, favourable or not, will be recorded without altering the predictions above.

Part II — Formula Compendium: Physics and Chemistry Formulas in v_e and H

Middle-rung law. At every epoch t:
$\hbar \cdot G = c^5 / (4\pi^2 \cdot v_e(t) \cdot H(t))$, equivalently $v_e(t) \cdot H(t) = v_P^2$, with $v_P = \sqrt{(c^5/\hbar G)}/2\pi$ the Planck (human-boundary) frequency.
v_e and H both change with cosmic time; their product is fixed, so ħ and G stay constant.

Axioms and definitions (1–7)

1. **The universe consists only of ether structures and ether motion.** Different ether structures are what humans call different kinds of matter. Humans are ether structures too.
2. **Basic ether structure:** the regular tetrahedron. Every other structure (all matter) is a knot of it.
3. **Ether space around every structure — flow through the knot:** the basic ether is the finest unit ($v_e \approx 4.3\times10^{102}$ Hz, 10^{79} times the proton frequency), so it passes through every structure. It enters along the thumb axis, passes through the knot, and leaves spiralling outward at light speed in the plane of the four fingers, which curl counterclockwise around the thumb (right-hand rule); the pattern resembles dipole field lines. A knot is a persistent pattern in the flow, like a whirlpool, not a container, so no ether is created or destroyed (continuity holds), and every knot is carried by the flow with the same acceleration (equivalence principle). A macroscopic body contains vast numbers of knots with random axes, so its gravity is isotropic. Every structure, without exception, carries such a surrounding ether space.
4. **Relations between structures:** forces (gravitational, electromagnetic, …) are field-pressure differences where the ether spaces of different structures overlap. Flow contributions add in quadrature ($v^2 = 2\Phi$ superposes linearly); a uniform ether produces no force.
5. **Light speed is the background; only deviations are measurable.** The measurable part around a mass is the deviation flow $\sqrt{(2GM/r)}$ (11.2 km/s at Earth's surface). An outward flow of this speed has fluid acceleration $v\cdot dv/dr = -GM/r^2$, identical to an inward flow, so the outward-spiral reading reproduces the same exterior (Schwarzschild) geometry and all solar-system tests.
6. **Elements:** ether structure n = the element with atomic number n (1 = H … 118 = Og); every substance is "ether structure number + frequency", with frequency $f = mc^2/h$.
7. **Cosmic evolution of the ether frequency and the exact coasting transition 9.90 Gyr ago (at cosmic age 3.91 Gyr):** by the middle-rung law $v_e(t)\cdot H(t) = v_P^2 = 8.715\times10^{84}$ Hz², the ether frequency rises as the universe ages while the expansion rate falls, keeping their product invariant (and keeping ħ and G constant within the lunar laser ranging bound $|\dot{G}/G| < 1.5\times10^{-13}$ yr⁻¹). The evolution proceeds in two phases: (i) **Early non-uniform phase (from the Big Bang to cosmic age 3.912 Gyr [3.641–3.950 Gyr across H_0 = 61.8–63.5])**: prior to galaxy stabilization, the universe undergoes hot radiation/matter evolution and intense hierarchical merging of smaller structural units (within this phase, $v_e = 10^{96}$ Hz [8 orders × 12 dimensions] and 1.51×10^{96} Hz correspond to exact cosmic ages of **1,866 yr** and **2,833 yr**, or 3,636 yr and 5,491 yr under pure H = 1/t extrapolation). (ii) **Late uniform coasting phase (from 9.900 Gyr ago [9.893–9.933 Gyr ago] to the present)**: at cosmic age **3.912 Gyr** (z_s = 1.686, range 1.669–1.817; at the acoustic-angle benchmark z_s = 1.721, age 3.837 Gyr and lookback time 9.911 Gyr, where $1+z_s = 2.721 \approx e$), smaller structural units complete their assembly into stable large-scale galaxies, ether cancellation reaches its steady state ($N^{-2} \approx 2.2\times10^{-120}$, 10 orders × 12 dimensions), and expansion locks into uniform coasting $H(z) = H_0(1+z)$. Because the z_s-dependence in $t_0 - t_s = z_s/[H_0(1+z_s)]$ cancels to first order against $H_0 \propto \sqrt{(1+z_s)}$, the lookback time to stabilization is tightly pinned across the entire H_0 = 61.8–63.5 km s⁻¹ Mpc⁻¹ band to **9.893–9.933 Gyr ago (midpoint 9.900 Gyr ago)**. Over the subsequent 9.900 Gyr, $v_e(t)$ grows linearly with cosmic time from 1.615×10^{102} Hz to 4.337×10^{102} Hz while H(t) declines inversely, maintaining $v_e(t)\cdot H(t) = 8.715\times10^{84}$ Hz².

Open point: whether the thumb axis corresponds to spin/magnetic moment. Reading (identity, not a discovery): a knot frequency $f = mc^2/h = c/\lambda_C$, i.e. the number of light-speed ether passages per second across the knot's own Compton wavelength.

1. Present values (H_0 = 61.8–63.5 km s⁻¹ Mpc⁻¹; central 62.0)

Quantity	Central (H_0 = 62.0)	Range (61.8–63.5)
Expansion rate H (universe rung)	2.009×10^{-18} s⁻¹	2.003×10^{-18} – 2.058×10^{-18} s⁻¹
Ether frequency $v_e = v_P^2/H$	4.337×10^{102} Hz	4.235×10^{102} – 4.351×10^{102} Hz
Planck (human-boundary) frequency v_P	2.952×10^{42} Hz	constant

Invariant product $v_e \cdot H = v_P^2$	8.715×10^{84} Hz²	constant
Rung ratio $N = v_P/H$	1.469×10^{60}	1.435×10^{60} – 1.474×10^{60}
Universe age ($H_0 t_0 = 0.876$)	13.82 Gyr	13.49–13.86 Gyr
Galaxy stabilization & coasting onset ($z_s = 1.686$)	9.900 Gyr ago (cosmic age 3.912 Gyr)	9.893–9.933 Gyr ago (age 3.641–3.950 Gyr)
Planck mass (human-boundary mass m_P)	21.76 μg (2.176×10^{-8} kg)	collective blind band baseline 21.8 ± 0.3 μg (21.5–22.1 μg)
Mid-rung frequency $\sqrt{(H\, v_P)}$	2.435 THz	2.43–2.46 THz

Numerically: $\hbar \cdot G = 7.039\times10^{-45}$ J·m³ kg⁻¹ s⁻² = $c^5 / (4\pi^2 \cdot 4.337\times10^{102} \cdot 2.009\times10^{-18})$.

2. Physics formulas — symbolic form (v_e, H vary with time) and present-value form

Formula	Standard	Ether form (time-dependent $v_e(t)$, H(t))	Present-value form ($v_e \cdot H = 8.71\times10^{84}$ Hz²)	Value
Mass–energy	$E = mc^2$	$E = mc^2$ (unchanged); knot frequency $f = mc^2/h$	$E = mc^2$	8.988×10^{16} J (m = 1 kg)
Planck relation	$E = h\nu$	$E = c^5\nu / (2\pi G\, v_e\, H)$	$E = c^5\nu / (2\pi G \cdot 4.34\times10^{102} \cdot 2.01\times10^{-18})$	6.626×10^{-19} J ($\nu = 10^{15}$ Hz)
de Broglie wavelength	$\lambda = h/p$	$\lambda = c^5 / (2\pi G\, v_e\, H\, p)$	$\lambda = c^5 / (2\pi G \cdot 8.71\times10^{84} \cdot p)$	7.274×10^{-10} m (electron, 10^6 m/s)
Newtonian gravity	$F = GMm/r^2$	$F = c^5\, M\, m / (4\pi^2 \hbar\, v_e\, H\, r^2)$	$F = c^5\, M\, m / (4\pi^2 \hbar \cdot 8.71\times10^{84} \cdot r^2)$	9.82 N (1 kg at Earth surface)
Ether inflow = escape speed	$v = \sqrt{(2GM/r)}$	$v = \sqrt{(c^5 M / (2\pi^2 \hbar\, v_e\, H\, r))}$	$v = \sqrt{(c^5 M / (2\pi^2 \hbar \cdot 8.71\times10^{84} \cdot r))}$	11,190 m/s (Earth surface)
Gravitational redshift	$\Delta f/f = GM/(rc^2)$	$\Delta f/f = f_M\, f_r / (v_e\, H)$, $f_M = Mc^2/h$, $f_r = c/2\pi r$	$\Delta f/f = f_M\, f_r / 8.71\times10^{84}$	6.961×10^{-10} (Earth surface)
Schwarzschild radius	$r_s = 2GM/c^2$	$r_s = c^3 M / (2\pi^2 \hbar\, v_e\, H)$	$r_s = c^3 M / (2\pi^2 \hbar \cdot 8.71\times10^{84})$	2954 m (Sun)
Planck length	$\ell_P = \sqrt{(\hbar G/c^3)}$	$\ell_P = c / (2\pi \sqrt{(v_e\, H)})$	$\ell_P = c / (2\pi \sqrt{(8.71\times10^{84})})$	1.616×10^{-35} m
Planck time	$t_P = \sqrt{(\hbar G/c^5)}$	$t_P = 1 / (2\pi \sqrt{(v_e\, H)})$	$t_P = 1 / (2\pi \sqrt{(8.71\times10^{84})})$	5.391×10^{-44} s
Planck mass	$m_P = \sqrt{(\hbar c/G)}$	$m_P = 2\pi \hbar \sqrt{(v_e\, H)} / c^2$	$m_P = 2\pi \hbar \sqrt{(8.71\times10^{84})} / c^2$	2.176×10^{-8} kg
Black-hole entropy	$S = k_B c^3 A / (4\hbar G)$	$S = \pi^2 k_B\, v_e\, H\, A / c^2$	$S = \pi^2 k_B \cdot 8.71\times10^{84} \cdot A / c^2$	1.321×10^{46} J/K (A = 1 m²)
Hawking temperature	$T = \hbar c^3 / (8\pi G M k_B)$	$T = \pi \hbar^2\, v_e\, H / (2c^2\, M\, k_B)$	$T = \pi \hbar^2 \cdot 8.71\times10^{84} / (2c^2\, M\, k_B)$	6.168×10^{-8} K (solar-mass black hole)
Fine-structure constant	$\alpha = e^2/(4\pi\varepsilon_0 \hbar c)$	$\alpha = \pi\, e^2\, G\, v_e\, H / (\varepsilon_0\, c^6)$	$\alpha = \pi\, e^2\, G \cdot 8.71\times10^{84} / (\varepsilon_0\, c^6)$	7.297×10^{-3} (dimensionless; e = electron charge)
Bohr radius	$a_0 = 4\pi\varepsilon_0 \hbar^2 / (m_e e^2)$	$a_0 = \varepsilon_0\, c^{10} / (4\pi^3\, G^2\, v_e^2\, H^2\, m_e\, e^2)$	$a_0 = \varepsilon_0\, c^{10} / (4\pi^3\, G^2\, (8.71\times10^{84})^2\, m_e\, e^2)$	5.292×10^{-11} m
Stefan–Boltzmann constant	$\sigma = \pi^2 k_B^4 / (60\hbar^3 c^2)$	$\sigma = \pi^2\, k_B^4\, (4\pi^2\, G\, v_e\, H)^3 / (60\, c^{17})$	$\sigma = \pi^2\, k_B^4\, (4\pi^2\, G \cdot 8.71\times10^{84})^3 / (60\, c^{17})$	5.670×10^{-8} W m⁻² K⁻⁴
Hubble law	$v = H\, d$	$v = H\, d$ (H = universe-rung frequency)	$v = 2.01\times10^{-18}$ s⁻¹ · d	62,000 m/s (d = 1 Mpc)
Normalization (Zhang 2026) on Hubble sphere	$4\pi^2 r^3 c^2 / (T^2 G h \nu) = 1$	$r = c/H$, $T = 2\pi/H \Rightarrow v = c^5/(hGH) = 2\pi\, v_e$	$v = 2\pi \cdot 4.34\times10^{102}$	2.725×10^{103} Hz
Temperature → frequency	$v = k_B T / h$	$v = 2\pi\, k_B\, T\, G\, v_e\, H / c^5$	$v = 2\pi\, k_B\, T\, G \cdot 8.71\times10^{84} / c^5$	6.459×10^{12} Hz (T = 310 K)
Two knots: gravity	$F = Gm_1m_2/r^2$	$F = c\, h\, f_1\, f_2 / (2\pi\, v_e\, H\, r^2)$	$F = c\, h\, f_1\, f_2 / (2\pi \cdot 8.71\times10^{84} \cdot r^2)$	6.674×10^{-11} N (two 1 kg, 1 m)

Electric / gravity ratio	$F_e/F_g = e^2/(4\pi\varepsilon_0 G m_1 m_2)$	$F_e/F_g = \alpha\, v_e\, H / (f_1\, f_2)$	$F_e/F_g = \alpha \cdot 8.71\times10^{84} / (f_1\, f_2)$	1.236×10^{36} (two protons)
Electric = gravity crossover	Stoney mass $\sqrt{\alpha}\, m_P$	$f = \sqrt{(\alpha\, v_e\, H)}$, $m = h\, f / c^2$	$f = \sqrt{(\alpha \cdot 8.71\times10^{84})}$	1.859×10^{-9} kg

Each ether form is an algebraic rewriting via the middle-rung law; the value column was checked to agree with the standard form to 10^{-9}. The physical content lies in the interpretation and in the dated falsifiable predictions (P1–P6.1), not in the numerical agreement.

3. Chemistry formulas (natural constant e appears only in exponentials and logarithms)

Formula	Form
Ideal gas	$PV = nRT$, $R = N_A k_B$
Boltzmann distribution	$n_2/n_1 = e^{-h\Delta\nu/k_B T}$
Arrhenius	$k = A\, e^{-E_a/RT}$
Nernst	$E = E^0 - (RT/zF) \ln Q$; $2.303\, k_B T/e = 61.5$ mV per decade at 310 K
Gibbs / equilibrium	$\Delta G^0 = -RT \ln K$, $K = e^{-\Delta G^0/RT}$; $\Delta G = -nFE$
Proton-motive force	$\Delta p = \Delta\psi - (2.303RT/F)\, \Delta pH$
Radioactive decay	$N = N_0\, e^{-\lambda t}$, $t_{1/2} = \ln 2/\lambda$
Photon energy	$E = hc/\lambda = c^6 / (2\pi\, G\, v_e\, H\, \lambda)$
Rydberg constant	$R_\infty = \pi^3\, m_e\, e^4\, G^3\, v_e^3\, H^3 / (\varepsilon_0^2\, c^{16})$
Bond energy → frequency	$\nu = \Delta E/h = 2\pi\, \Delta E\, G\, v_e\, H / c^5$

4. Elements as ether structures 1–118

Structure number = atomic number. Frequency $f = mc^2/h$. Ladder position $k = \log(m/m_U)/\log(m_P/m_U)$, with $m_U = hH/c^2$ and m_P the Planck mass. Ether form: $f = H^{1-k/2} \cdot v_e^{k/2}$ (exact). Values for $H_0 = 62.0$.

Z	Symbol	Atomic mass (u)	f (Hz)	k	Ether form
1	H	1.008	2.270×10^{23}	0.6823	$H^{0.6588}.v_e^{0.3412}$
2	He	4.0026	9.015×10^{23}	0.6923	$H^{0.6539}.v_e^{0.3461}$
3	Li	6.94	1.563×10^{24}	0.6962	$H^{0.6519}.v_e^{0.3481}$
4	Be	9.0122	2.030×10^{24}	0.6981	$H^{0.6509}.v_e^{0.3491}$
5	B	10.81	2.435×10^{24}	0.6994	$H^{0.6503}.v_e^{0.3497}$
6	C	12.011	2.705×10^{24}	0.7002	$H^{0.6499}.v_e^{0.3501}$
7	N	14.007	3.155×10^{24}	0.7013	$H^{0.6493}.v_e^{0.3507}$
8	O	15.999	3.604×10^{24}	0.7023	$H^{0.6489}.v_e^{0.3511}$
9	F	18.998	4.279×10^{24}	0.7035	$H^{0.6482}.v_e^{0.3518}$
10	Ne	20.18	4.545×10^{24}	0.7039	$H^{0.6480}.v_e^{0.3520}$
11	Na	22.99	5.178×10^{24}	0.7049	$H^{0.6476}.v_e^{0.3524}$
12	Mg	24.305	5.474×10^{24}	0.7053	$H^{0.6474}.v_e^{0.3526}$
13	Al	26.982	6.077×10^{24}	0.7060	$H^{0.6470}.v_e^{0.3530}$
14	Si	28.085	6.326×10^{24}	0.7063	$H^{0.6468}.v_e^{0.3532}$
15	P	30.974	6.976×10^{24}	0.7070	$H^{0.6465}.v_e^{0.3535}$
16	S	32.06	7.221×10^{24}	0.7073	$H^{0.6464}.v_e^{0.3536}$
17	Cl	35.45	7.985×10^{24}	0.7080	$H^{0.6460}.v_e^{0.3540}$
18	Ar	39.95	8.998×10^{24}	0.7089	$H^{0.6456}.v_e^{0.3544}$
19	K	39.098	8.806×10^{24}	0.7087	$H^{0.6456}.v_e^{0.3544}$
20	Ca	40.078	9.027×10^{24}	0.7089	$H^{0.6455}.v_e^{0.3545}$
21	Sc	44.956	1.013×10^{25}	0.7097	$H^{0.6451}.v_e^{0.3549}$
22	Ti	47.867	1.078×10^{25}	0.7102	$H^{0.6449}.v_e^{0.3551}$
23	V	50.942	1.147×10^{25}	0.7106	$H^{0.6447}.v_e^{0.3553}$
24	Cr	51.996	1.171×10^{25}	0.7108	$H^{0.6446}.v_e^{0.3554}$
25	Mn	54.938	1.237×10^{25}	0.7112	$H^{0.6444}.v_e^{0.3556}$
26	Fe	55.845	1.258×10^{25}	0.7113	$H^{0.6444}.v_e^{0.3556}$
27	Co	58.933	1.327×10^{25}	0.7117	$H^{0.6442}.v_e^{0.3558}$
28	Ni	58.693	1.322×10^{25}	0.7117	$H^{0.6442}.v_e^{0.3558}$
29	Cu	63.546	1.431×10^{25}	0.7122	$H^{0.6439}.v_e^{0.3561}$
30	Zn	65.38	1.473×10^{25}	0.7124	$H^{0.6438}.v_e^{0.3562}$
31	Ga	69.723	1.570×10^{25}	0.7129	$H^{0.6436}.v_e^{0.3564}$
32	Ge	72.63	1.636×10^{25}	0.7132	$H^{0.6434}.v_e^{0.3566}$
33	As	74.922	1.688×10^{25}	0.7134	$H^{0.6433}.v_e^{0.3567}$
34	Se	78.971	1.779×10^{25}	0.7138	$H^{0.6431}.v_e^{0.3569}$
35	Br	79.904	1.800×10^{25}	0.7139	$H^{0.6431}.v_e^{0.3569}$

36	Kr	83.798	1.887×10^{25}	0.7142	$H^{0.6429}.v_e^{0.3571}$
37	Rb	85.468	1.925×10^{25}	0.7144	$H^{0.6428}.v_e^{0.3572}$
38	Sr	87.62	1.974×10^{25}	0.7145	$H^{0.6427}.v_e^{0.3573}$
39	Y	88.906	2.002×10^{25}	0.7147	$H^{0.6427}.v_e^{0.3573}$
40	Zr	91.224	2.055×10^{25}	0.7148	$H^{0.6426}.v_e^{0.3574}$
41	Nb	92.906	2.093×10^{25}	0.7150	$H^{0.6425}.v_e^{0.3575}$
42	Mo	95.95	2.161×10^{25}	0.7152	$H^{0.6424}.v_e^{0.3576}$
43	Tc	98	2.207×10^{25}	0.7154	$H^{0.6423}.v_e^{0.3577}$
44	Ru	101.07	2.276×10^{25}	0.7156	$H^{0.6422}.v_e^{0.3578}$
45	Rh	102.91	2.318×10^{25}	0.7157	$H^{0.6421}.v_e^{0.3579}$
46	Pd	106.42	2.397×10^{25}	0.7159	$H^{0.6420}.v_e^{0.3580}$
47	Ag	107.87	2.430×10^{25}	0.7160	$H^{0.6420}.v_e^{0.3580}$
48	Cd	112.41	2.532×10^{25}	0.7163	$H^{0.6418}.v_e^{0.3582}$
49	In	114.82	2.586×10^{25}	0.7165	$H^{0.6418}.v_e^{0.3582}$
50	Sn	118.71	2.674×10^{25}	0.7167	$H^{0.6416}.v_e^{0.3584}$
51	Sb	121.76	2.742×10^{25}	0.7169	$H^{0.6415}.v_e^{0.3585}$
52	Te	127.6	2.874×10^{25}	0.7173	$H^{0.6414}.v_e^{0.3586}$
53	I	126.9	2.858×10^{25}	0.7172	$H^{0.6414}.v_e^{0.3586}$
54	Xe	131.29	2.957×10^{25}	0.7175	$H^{0.6413}.v_e^{0.3587}$
55	Cs	132.91	2.994×10^{25}	0.7176	$H^{0.6412}.v_e^{0.3588}$
56	Ba	137.33	3.093×10^{25}	0.7178	$H^{0.6411}.v_e^{0.3589}$
57	La	138.91	3.129×10^{25}	0.7179	$H^{0.6411}.v_e^{0.3589}$
58	Ce	140.12	3.156×10^{25}	0.7179	$H^{0.6410}.v_e^{0.3590}$
59	Pr	140.91	3.174×10^{25}	0.7180	$H^{0.6410}.v_e^{0.3590}$
60	Nd	144.24	3.249×10^{25}	0.7181	$H^{0.6409}.v_e^{0.3591}$
61	Pm	145	3.266×10^{25}	0.7182	$H^{0.6409}.v_e^{0.3591}$
62	Sm	150.36	3.387×10^{25}	0.7184	$H^{0.6408}.v_e^{0.3592}$
63	Eu	151.96	3.423×10^{25}	0.7185	$H^{0.6407}.v_e^{0.3593}$
64	Gd	157.25	3.542×10^{25}	0.7188	$H^{0.6406}.v_e^{0.3594}$
65	Tb	158.93	3.580×10^{25}	0.7188	$H^{0.6406}.v_e^{0.3594}$
66	Dy	162.5	3.660×10^{25}	0.7190	$H^{0.6405}.v_e^{0.3595}$
67	Ho	164.93	3.715×10^{25}	0.7191	$H^{0.6404}.v_e^{0.3596}$
68	Er	167.26	3.767×10^{25}	0.7192	$H^{0.6404}.v_e^{0.3596}$
69	Tm	168.93	3.805×10^{25}	0.7193	$H^{0.6404}.v_e^{0.3596}$
70	Yb	173.05	3.898×10^{25}	0.7195	$H^{0.6403}.v_e^{0.3597}$
71	Lu	174.97	3.941×10^{25}	0.7195	$H^{0.6402}.v_e^{0.3598}$
72	Hf	178.49	4.020×10^{25}	0.7197	$H^{0.6402}.v_e^{0.3598}$
73	Ta	180.95	4.076×10^{25}	0.7198	$H^{0.6401}.v_e^{0.3599}$
74	W	183.84	4.141×10^{25}	0.7199	$H^{0.6401}.v_e^{0.3599}$

75	Re	186.21	4.194×10^{25}	0.7200	$H^{0.6400}.\nu_e{}^{0.3600}$
76	Os	190.23	4.285×10^{25}	0.7201	$H^{0.6399}.\nu_e{}^{0.3601}$
77	Ir	192.22	4.329×10^{25}	0.7202	$H^{0.6399}.\nu_e{}^{0.3601}$
78	Pt	195.08	4.394×10^{25}	0.7203	$H^{0.6398}.\nu_e{}^{0.3602}$
79	Au	196.97	4.436×10^{25}	0.7204	$H^{0.6398}.\nu_e{}^{0.3602}$
80	Hg	200.59	4.518×10^{25}	0.7205	$H^{0.6397}.\nu_e{}^{0.3603}$
81	Tl	204.38	4.603×10^{25}	0.7207	$H^{0.6397}.\nu_e{}^{0.3603}$
82	Pb	207.2	4.667×10^{25}	0.7208	$H^{0.6396}.\nu_e{}^{0.3604}$
83	Bi	208.98	4.707×10^{25}	0.7208	$H^{0.6396}.\nu_e{}^{0.3604}$
84	Po	209	4.707×10^{25}	0.7208	$H^{0.6396}.\nu_e{}^{0.3604}$
85	At	210	4.730×10^{25}	0.7209	$H^{0.6396}.\nu_e{}^{0.3604}$
86	Rn	222	5.000×10^{25}	0.7213	$H^{0.6394}.\nu_e{}^{0.3606}$
87	Fr	223	5.023×10^{25}	0.7213	$H^{0.6394}.\nu_e{}^{0.3606}$
88	Ra	226	5.090×10^{25}	0.7214	$H^{0.6393}.\nu_e{}^{0.3607}$
89	Ac	227	5.113×10^{25}	0.7214	$H^{0.6393}.\nu_e{}^{0.3607}$
90	Th	232.04	5.226×10^{25}	0.7216	$H^{0.6392}.\nu_e{}^{0.3608}$
91	Pa	231.04	5.204×10^{25}	0.7215	$H^{0.6392}.\nu_e{}^{0.3608}$
92	U	238.03	5.361×10^{25}	0.7218	$H^{0.6391}.\nu_e{}^{0.3609}$
93	Np	237	5.338×10^{25}	0.7217	$H^{0.6391}.\nu_e{}^{0.3609}$
94	Pu	244	5.496×10^{25}	0.7219	$H^{0.6390}.\nu_e{}^{0.3610}$
95	Am	243	5.473×10^{25}	0.7219	$H^{0.6390}.\nu_e{}^{0.3610}$
96	Cm	247	5.563×10^{25}	0.7220	$H^{0.6390}.\nu_e{}^{0.3610}$
97	Bk	247	5.563×10^{25}	0.7220	$H^{0.6390}.\nu_e{}^{0.3610}$
98	Cf	251	5.653×10^{25}	0.7221	$H^{0.6389}.\nu_e{}^{0.3611}$
99	Es	252	5.676×10^{25}	0.7222	$H^{0.6389}.\nu_e{}^{0.3611}$
100	Fm	257	5.789×10^{25}	0.7223	$H^{0.6388}.\nu_e{}^{0.3612}$
101	Md	258	5.811×10^{25}	0.7223	$H^{0.6388}.\nu_e{}^{0.3612}$
102	No	259	5.834×10^{25}	0.7224	$H^{0.6388}.\nu_e{}^{0.3612}$
103	Lr	266	5.991×10^{25}	0.7226	$H^{0.6387}.\nu_e{}^{0.3613}$
104	Rf	267	6.014×10^{25}	0.7226	$H^{0.6387}.\nu_e{}^{0.3613}$
105	Db	268	6.036×10^{25}	0.7226	$H^{0.6387}.\nu_e{}^{0.3613}$
106	Sg	269	6.059×10^{25}	0.7226	$H^{0.6387}.\nu_e{}^{0.3613}$
107	Bh	270	6.081×10^{25}	0.7227	$H^{0.6387}.\nu_e{}^{0.3613}$
108	Hs	269	6.059×10^{25}	0.7226	$H^{0.6387}.\nu_e{}^{0.3613}$
109	Mt	278	6.262×10^{25}	0.7229	$H^{0.6386}.\nu_e{}^{0.3614}$
110	Ds	281	6.329×10^{25}	0.7230	$H^{0.6385}.\nu_e{}^{0.3615}$
111	Rg	282	6.352×10^{25}	0.7230	$H^{0.6385}.\nu_e{}^{0.3615}$
112	Cn	285	6.419×10^{25}	0.7231	$H^{0.6385}.\nu_e{}^{0.3615}$
113	Nh	286	6.442×10^{25}	0.7231	$H^{0.6385}.\nu_e{}^{0.3615}$

114	Fl	289	6.509×10^{25}	0.7232	$H^{0.6384}.v_e{}^{0.3616}$
115	Mc	290	6.532×10^{25}	0.7232	$H^{0.6384}.v_e{}^{0.3616}$
116	Lv	293	6.599×10^{25}	0.7233	$H^{0.6384}.v_e{}^{0.3616}$
117	Ts	294	6.622×10^{25}	0.7233	$H^{0.6384}.v_e{}^{0.3616}$
118	Og	294	6.622×10^{25}	0.7233	$H^{0.6384}.v_e{}^{0.3616}$

Part III — Addendum (12 November 2026): Mother-Equation Interface, Observation as Scanning Frequency, and an Interpretation of Parity Violation

This part is written by the first author. It changes none of the predictions or numbers in Parts I and II; the locked content of P1–P6.1 is unchanged. The Z-Model (mother-equation framework) is the first author's extension and does not represent the second author's views. Each item is graded: *rewrite* (algebraically equivalent to standard formulas, no new prediction), *derivation* (follows from the axioms, untested), *interpretation* (no numerical prediction).

III.1 Interface between the mother equation and the normalization equation

$$4\pi^2 \cdot r^3 \cdot (r_{lock}/r)^{\delta} \cdot (c^2 - v^2) \,/\, [\, T^2 \cdot E_{src} \cdot \mathcal{Z}_{eff}(r, v)\,] = 1$$

For $\delta \to 0$, $v = 0$, $E_{src} = h\nu$ and $\mathcal{Z}_{eff} \to \mathcal{Z}_0 = G$, the mother equation reduces exactly to the static normalization equation $4\pi^2 r^3 c^2/(T^2 G h\nu) = 1$ written by Zhang Xiangqian on 3 March 2026 (rewrite). The three added factors each carry one role: $(r_{lock}/r)^{\delta}$ carries the dimensional deficit of space at very small scales; $c^2 - v^2$ carries motion and observation, where v is the same quantity as $v^2 = 2\Phi$ in the observational-collective mechanism of Part I, and $v = 0$ is the static version; $\mathcal{Z}_{eff}$ carries the effective stiffness of space. The source energy E_{src} is two-state: $h\nu$ in the quantum state, Mc^2 in the gravitational state.

Base impedance in ether form: $\mathcal{Z}_0 = G = c^5/(4\pi^2 \hbar\, v_A\, H)$ (rewrite). Effective impedance $\mathcal{Z}_{eff}(r, v) = \mathcal{Z}_0[1 + \mathcal{C}]$, with $\mathcal{C}(r, v) = \mathcal{A}\,(r_{lock}/r)^{1/6}\,(v/v_{lock})^2$. An impedance larger by $(1+\mathcal{C})$ is equivalent to a local $v_A H$ smaller by $(1+\mathcal{C})$.

Provenance of parameters (stated plainly):

- **$\delta = 1/6$**: the topological deficit $\Delta N = -1$ shared over $N = 6$-fold symmetry, $\alpha = |\Delta N|/N = 1/6$; geometrically, a full turn of 360° missing 60°, leaving an effective closure angle of 300°. $\Delta N = -1$ and $N = 6$ are axiomatic inputs, not group-theoretic results. δ runs with $\mathcal{C}$, $\delta = (1/6)\cdot\mathcal{C}/(1+\mathcal{C})$; at macroscopic scales $\mathcal{C} \ll 1$ and $\delta \to 0$; the effective dimension $D_{eff} = 3 - \delta$ has the asymptotic limit 17/6.
- **$T_{lock} = 232$ as**, $r_{lock} = c\cdot T_{lock} \approx 69.6$ nm, $\nu_{lock} = 1/T_{lock} \approx 4.31\times10^{15}$ Hz (photon energy 17.8 eV). The 232 as value is taken from Jiang et al., Phys. Rev. Lett. 133, 163201 (2024), a time-dependent Schrödinger-equation calculation of helium ionization in a strong extreme-ultraviolet field: when the remaining electron ends in the lower-energy state, the ejected electron's "birth time" is on average about 232 as later. It is an average that depends on the atom and the laser configuration; promoting it to a cross-scale constant is a posterior insertion and cannot at present be derived from other measurable quantities.
- **Macroscopic null test**: for the Earth's orbit, $\nu_{orbit} \approx 3.17\times10^{-8}$ Hz, $(r_{lock}/r)^{1/6} \approx 8.8\times10^{-4}$, $\mathcal{C} \approx 4.8\times10^{-50}\,\mathcal{A}$; Kepler's law is unaffected.
- **Meaning of v**: v in $\mathcal{C}$ is a motion or probe frequency (orbital frequency, illuminating light), not the body's total frequency mc^2/h. Inserting the latter would make the Earth term $(v/v_{lock})^2 \approx 3.5\times10^{118}$, contradicting Kepler's law. The total frequency governs the Planck gate (III.2).

III.2 Observation as scanning frequency: an optical reading of the Planck gate

Observation works by light going out and coming back. To resolve the quantum wave of a mass m, the probe photon's reduced wavelength must shrink to $\lambda\!\!\!^{-} = \hbar/(mc)$; that photon then carries energy exactly mc^2 and frequency exactly the body's own total frequency $f = mc^2/h$ (rewrite). At that moment the observer's scanning frequency and the body's total frequency are the same number. The photon's own gravitational radius is $r_\gamma = 2GE/c^4 = 2Gm/c^2$, and the ratio $\lambda\!\!\!^{-}/r_\gamma = (m_P/m)^2/2$ is about 237 at 1 μg and 1/2 at 21.76 μg ($\lambda\!\!\!^{-} = \ell_P = 1.616\times10^{-35}$ m, $r_\gamma = 2\ell_P$). The light able to resolve the body's wave has already fallen inside its own gravitational radius, so the signal cannot return; the body and its surroundings stay visible in lower-frequency light, and only its wave becomes unreadable. This is the argument of Mead (1964) and of the generalized uncertainty principle; in ether form the gate frequency is $\nu_P = \sqrt{(\nu_A H)} = 2.952\times10^{42}$ Hz, the geometric mean of the ether base frequency and the expansion rate (rewrite).

Limits: the crossing $\lambda\!\!\!^{-} = r_\gamma$ lies at $m_P/\sqrt{2} \approx 15.4$ μg with $\hbar$ and at $\sqrt{\pi}\cdot m_P \approx 38.6$ μg with h; 21.76 μg lies between. The argument fixes only the order of magnitude of the gate; the ±0.3 μg width of the P6.1 window does not follow from it.

III.3 New derived prediction P6.2: the Planck gate shifts with probe-light frequency (registered 12 November 2026; not part of the 5 November locked content)

Replacing G with $\mathcal{Z}_{eff}$: $v_{P,eff} = v_P/\sqrt{(1+\mathcal{C})}$, $m_{P,eff} = m_P/\sqrt{(1+\mathcal{C})}$ (derivation). Under the framework's μ-degeneracy, $m\cdot\mathcal{Z}_{eff}$ is invariant, the field-pressure difference is unchanged, and the read-out mass decreases in proportion. For a 21.76 μg sample of radius 0.173 mm, the geometric factor $(r_{lock}/r)^{1/6} = 0.272$:

Probe light	$\mathcal{C}/\mathcal{A}$	Gate position for $\mathcal{A} = 1$
Visible, 2.0 eV	0.0034	21.73 μg
Ultraviolet, 10 eV	0.085	20.89 μg
Lock light, 17.8 eV (69.6 nm)	0.272	19.30 μg

Under visible light the gate moves by about 0.17%, inside the 21.8 ± 0.3 μg window of P6.1. Criterion: provided the P6.1 blind band has been observed, scan the same batch of samples in the same apparatus with visible light and with 17.8 eV lock light as the probe; if the band centre does not move toward lighter masses under lock light (shift below one third of the window width, or significance below 3σ), P6.2 is falsified; if it moves, the shift $\Delta m/m = 1 - 1/\sqrt{(1 + 0.272\mathcal{A})}$ yields $\mathcal{A}$. Note: 17.8 eV photons penetrate only nanometres to micrometres into air and most materials, so the experiment must be done in vacuum with surface-scattering read-out.

III.4 Interpretation: parity violation and the gap that makes observation possible (no numerical prediction)

By the axioms, observation is the field-pressure coupling produced when the ether spaces of the observed body and of the observational collective overlap. If the universe were perfectly symmetric and conserved at every level, the field-pressure difference would vanish everywhere ($\Delta\Phi = 0$): no gradient, no energy conversion, no signal, nothing to read out — like a closed room made entirely of mirrors, with nothing to compare against; an eye cannot see the eye. On this reading, the topological deficit $\Delta N = -1$ and the breaking of parity in the weak interaction are the small asymmetry that lets observation happen: the gap leaves room for an observer to stand in, separating subject from object, the mirror from what it reflects. The disappearance of observation at the Planck gate has the same shape: the scanning frequency, the probe light and the body's total frequency coincide at ν_P, leaving no readable difference between observer and observed.

Facts and open question: in 1957 Wu showed with cobalt-60 β decay that electrons are emitted preferentially opposite to the nuclear spin, so parity is violated in the weak interaction. The strong and electromagnetic interactions conserve parity, and the parity violation measured in atoms also comes from the weak neutral current. For this interpretation to stand, it must explain why the gap of observation falls on the weak interaction alone. This section is not a test basis and changes no prediction.